

American Samoa Model Checks

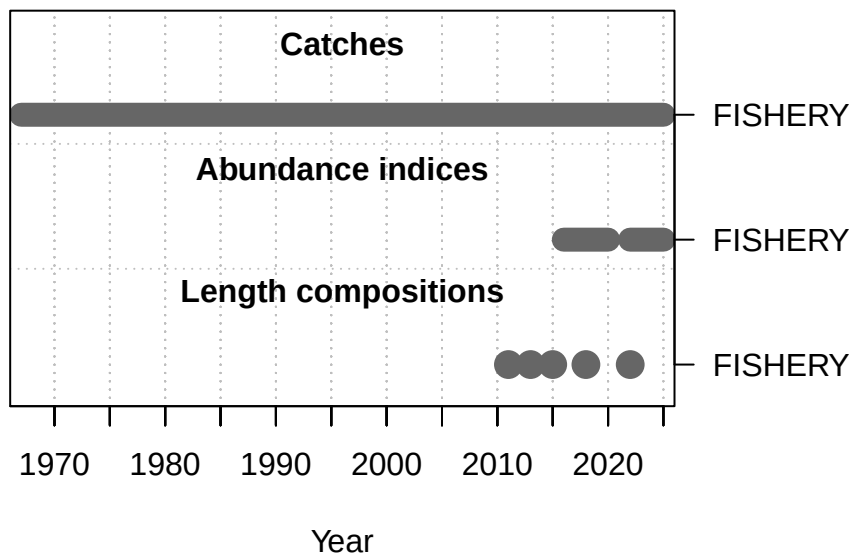
Marc Nadon and Meg Oshima

2026-04-06

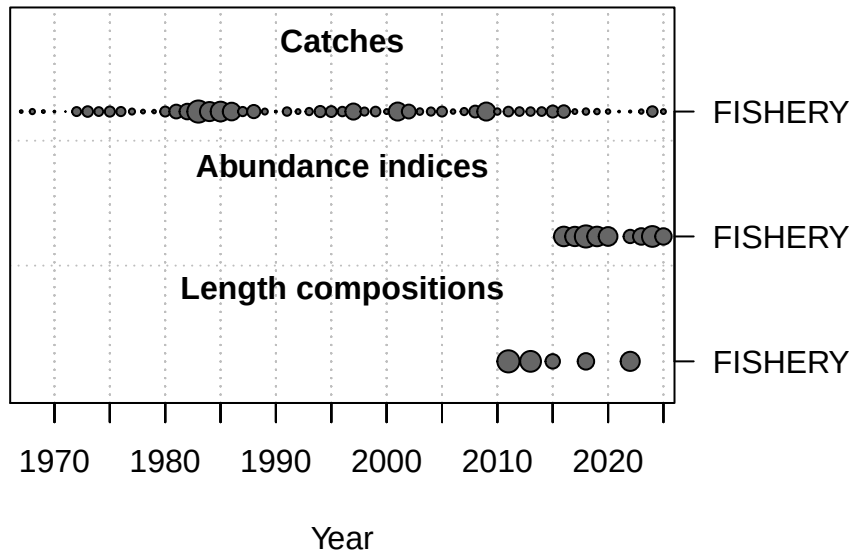
This is a summary report for the PRFL base model run.

Model Output

Input Data



Fleet	RMSE.perc	Nobs
FISHERY	55.6	9
Combined	55.6	9



Convergence Check

```

Converged      MaxGrad
1      TRUE 5.25778e-06
```

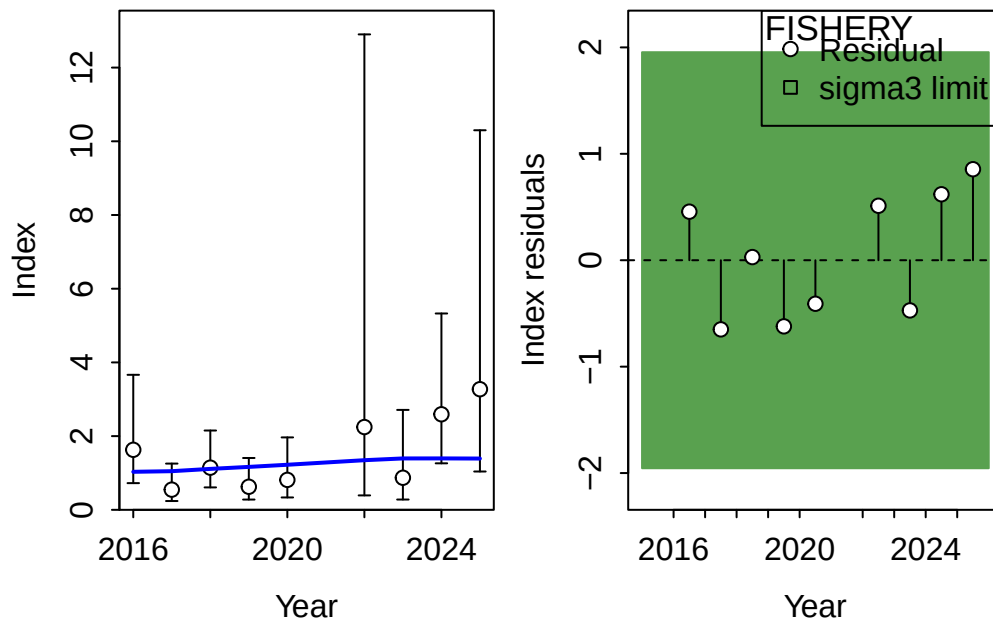
```

[1] "1 NOTE:  Max data length bin: 48  < max pop len bins: 53; so will accumulate larger pop
[2] "N warnings: 1"
```

Fleet	RMSE.perc	Nobs
FISHERY	4	5
Combined	4	5

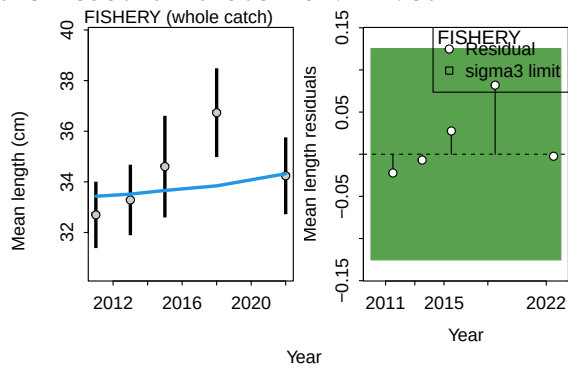
Fit to Model

CPUE

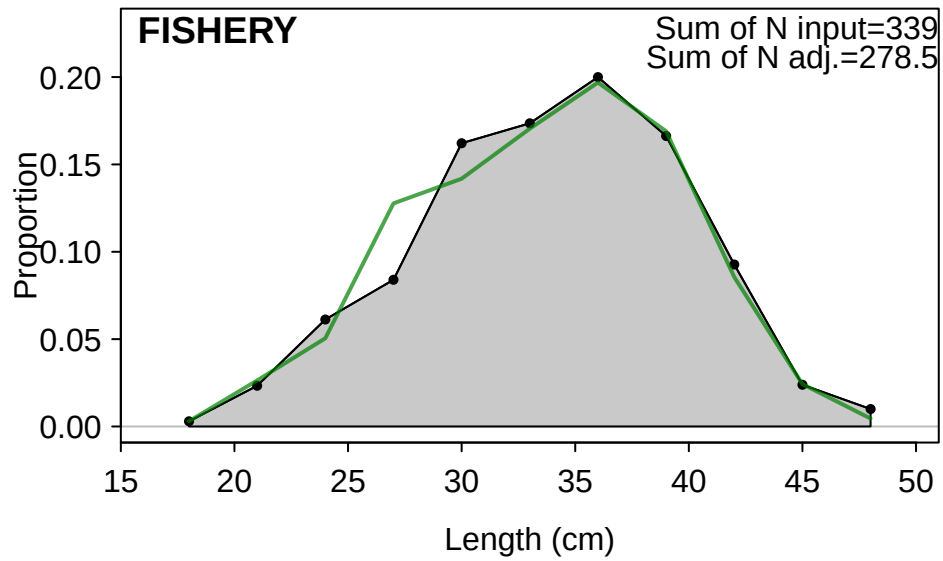


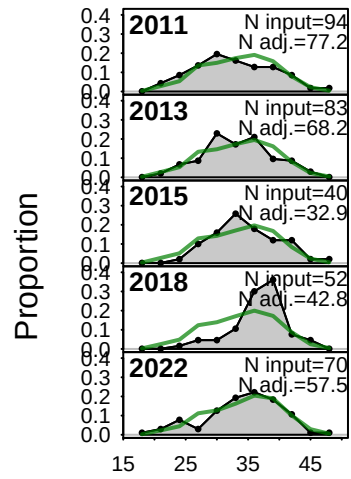
w lo hi
0.3423936 0.2029632 5.7241730

Index runs.p test sigma3.lo sigma3.hi type
1 FISHERY 0.331 Passed -0.1253339 0.1253339 len



Length Comp



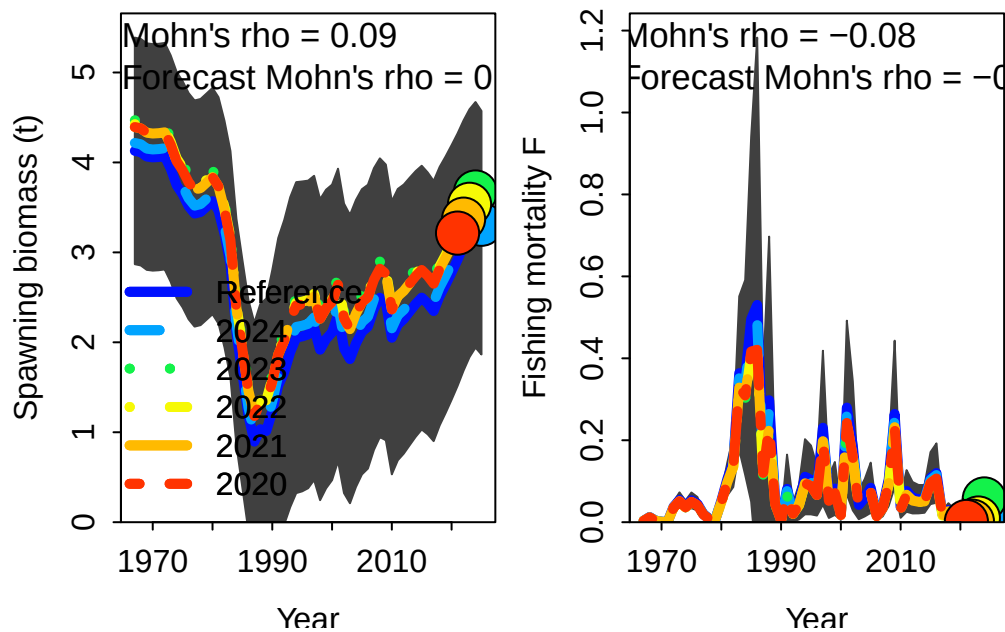


Length (cm)

Retrospective and Hindcasting

Retrospective

Mohn's Rho stats, including one step ahead forecasts:



Mohn's Rho stats, including one step ahead forecasts:

	type	peel	Rho	ForecastRho
1	F	2024	-0.02834875	-0.02821705
2	F	2023	-0.10109643	-0.10041465
3	F	2022	-0.09299765	-0.08947195
4	F	2021	-0.08840601	-0.08398570
5	F	2020	-0.09322314	-0.08840601
6	F	Combined	-0.08081440	-0.07809907

Hindcasting

Plotting Hindcast Cross-Validation (one-step-ahead)

Computing MASE with only 4 of 5 prediction residuals for Index FISHERY

Warning: Unequal spacing of naive predictions residuals may influence the interpretation of

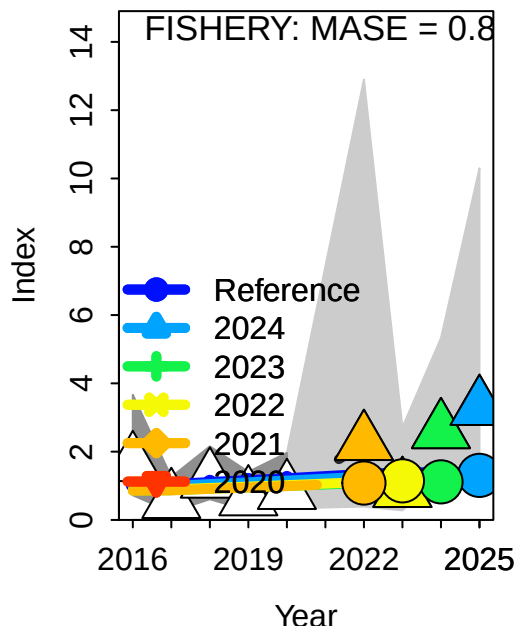
MASE stats by Index:

Plotting Hindcast Cross-Validation (one-step-ahead)

No observations in evaluation years to compute prediction residuals for Index FISHERY

MASE stats by Index:

	Index	Season	MASE	MAE.PR	MAE.base	MASE.adj	n.eval
1	FISHERY	1	NA	NA	NA	NA	0



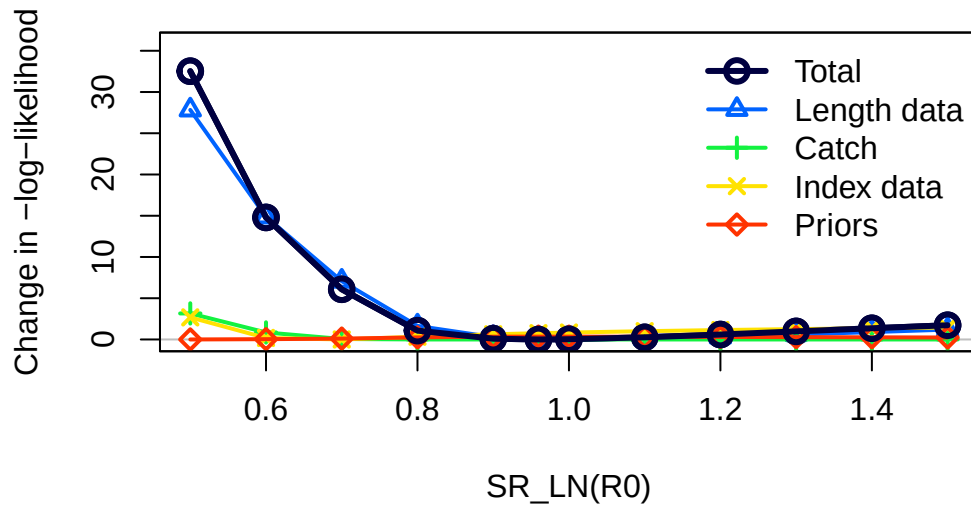
Recruitment Deviations

Likelihood Profile

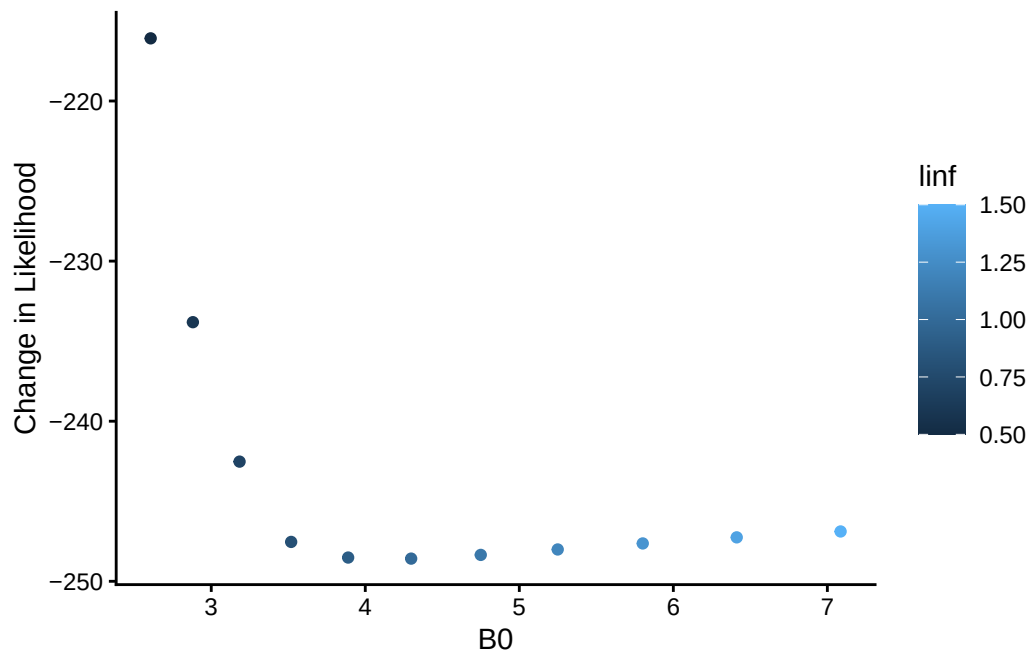
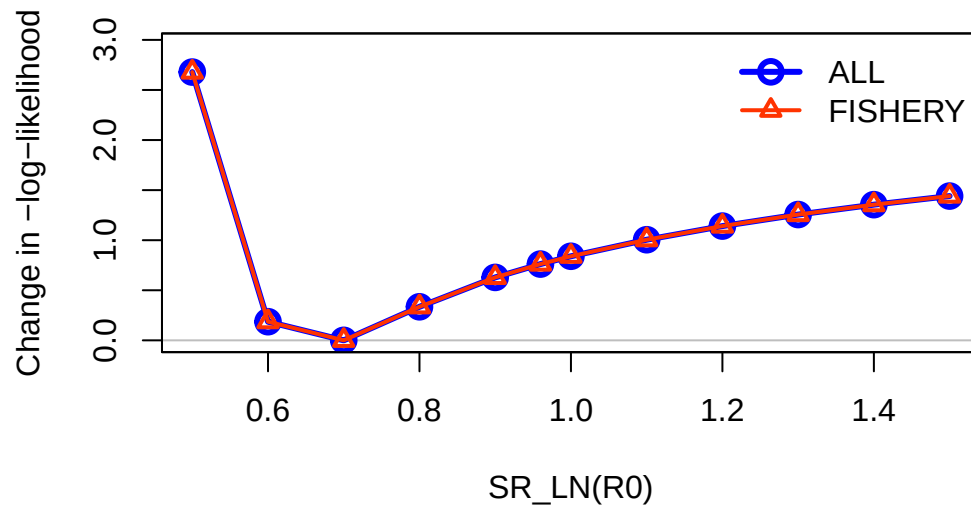
[1] "SR_LN(R0)"

	frac_change	include	label
TOTAL	1.0000	TRUE	Total
Catch	0.0971	TRUE	Catch
Equil_catch	0.0000	FALSE	Equilibrium catch
Survey	0.0824	TRUE	Index data
Length_comp	0.8557	TRUE	Length data
Recruitment	0.0000	FALSE	Recruitment

InitEQ_Regime	0.0000	FALSE	Initital equilibrium recruitment
Forecast_Recruitment	0.0000	FALSE	Forecast recruitment
Parm_priors	0.0106	TRUE	Priors
Parm_softbounds	0.0003	FALSE	Soft bounds
Parm_devs	0.0000	FALSE	Parameter deviations
Crash_Pen	0.0000	FALSE	Crash penalty

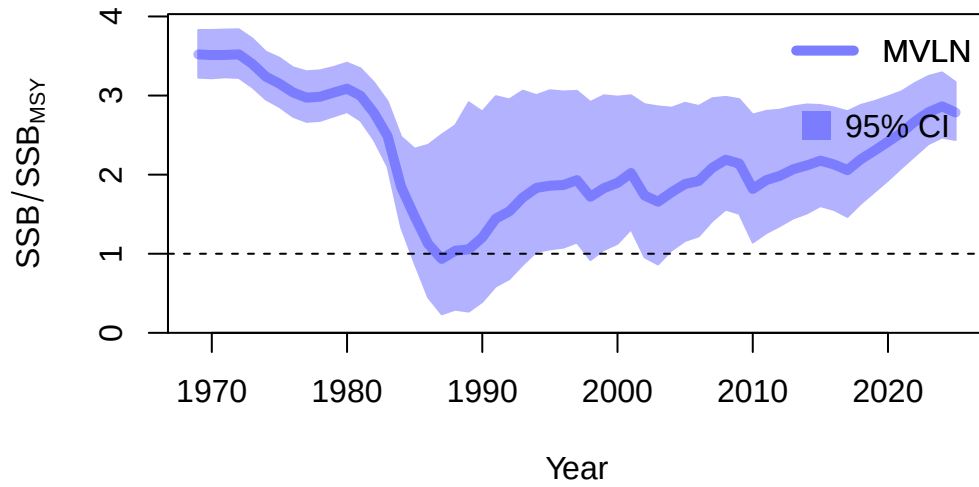
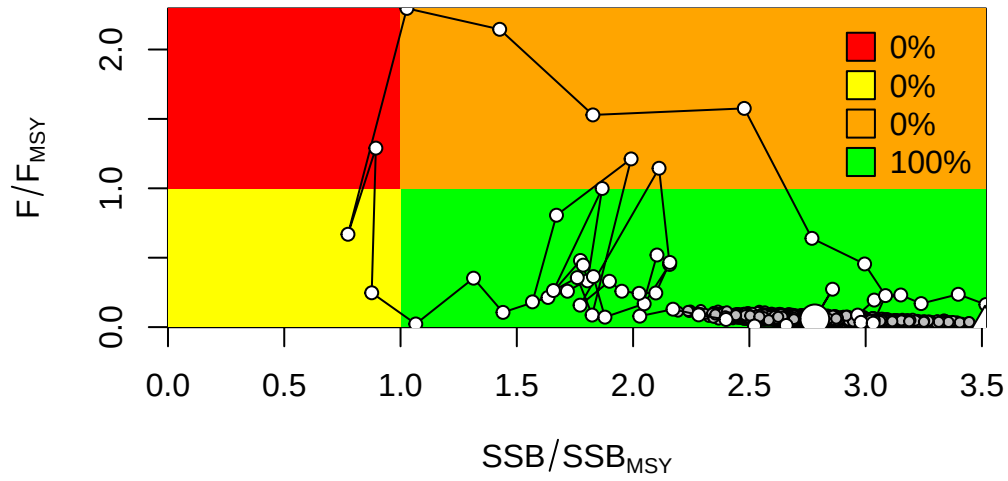


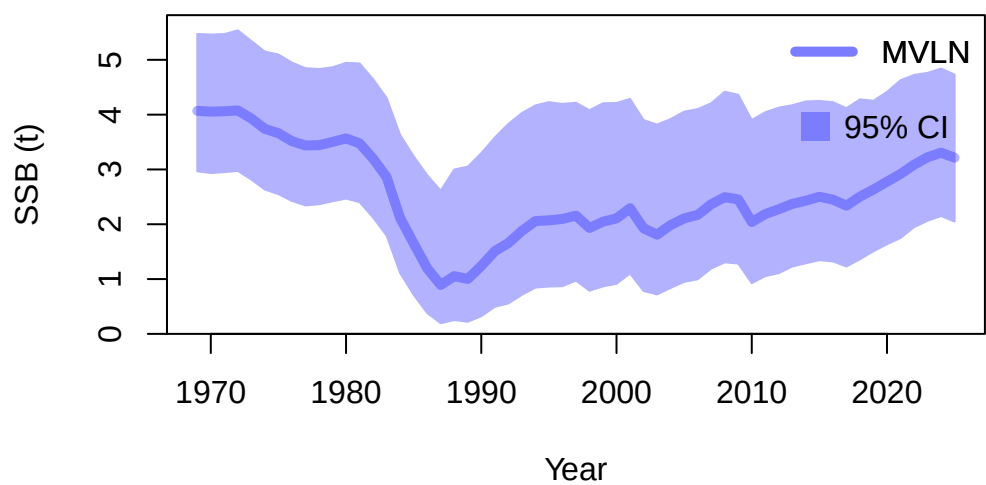
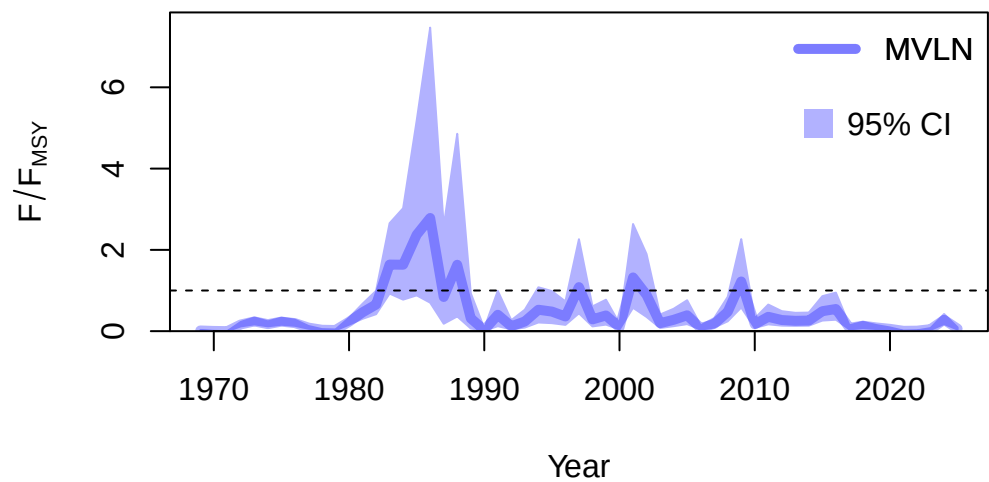
Changes in survey likelihood by fleet

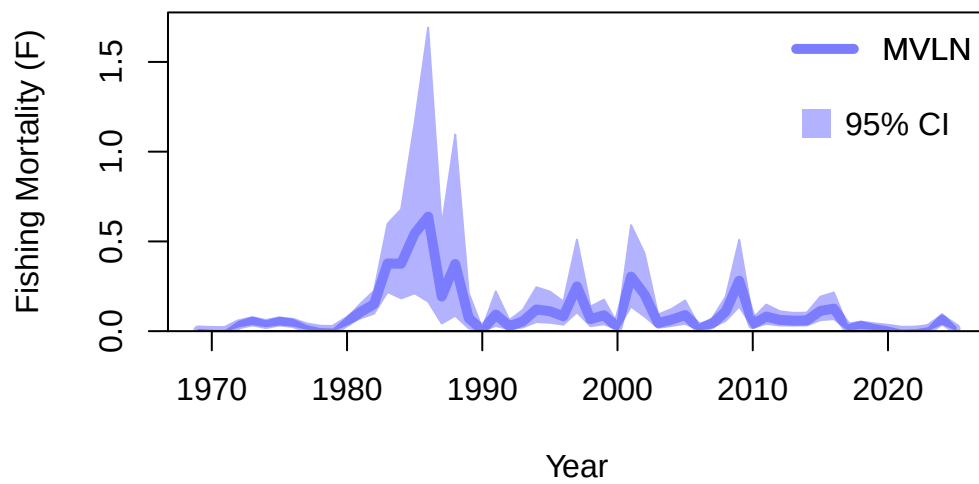


Management Quantities

starter.sso with Bratio: SSB/SSB_{MSY} and F: _abs_F

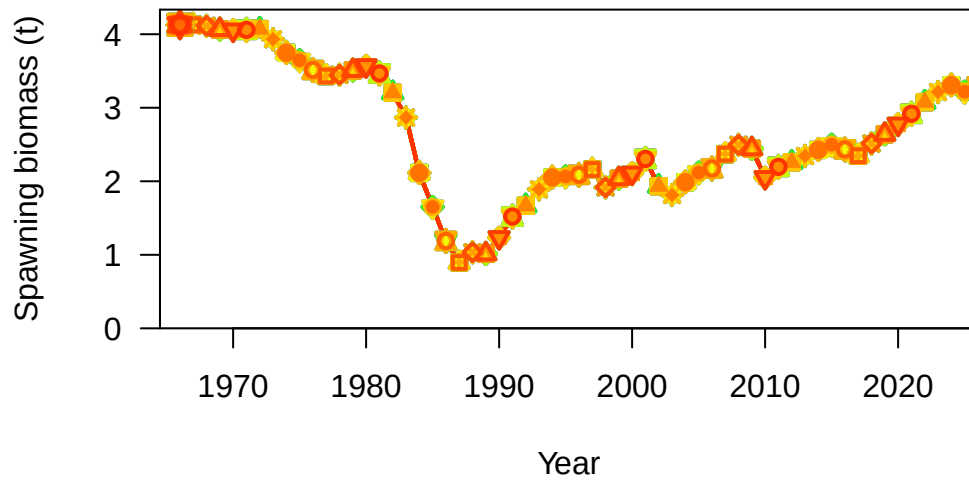
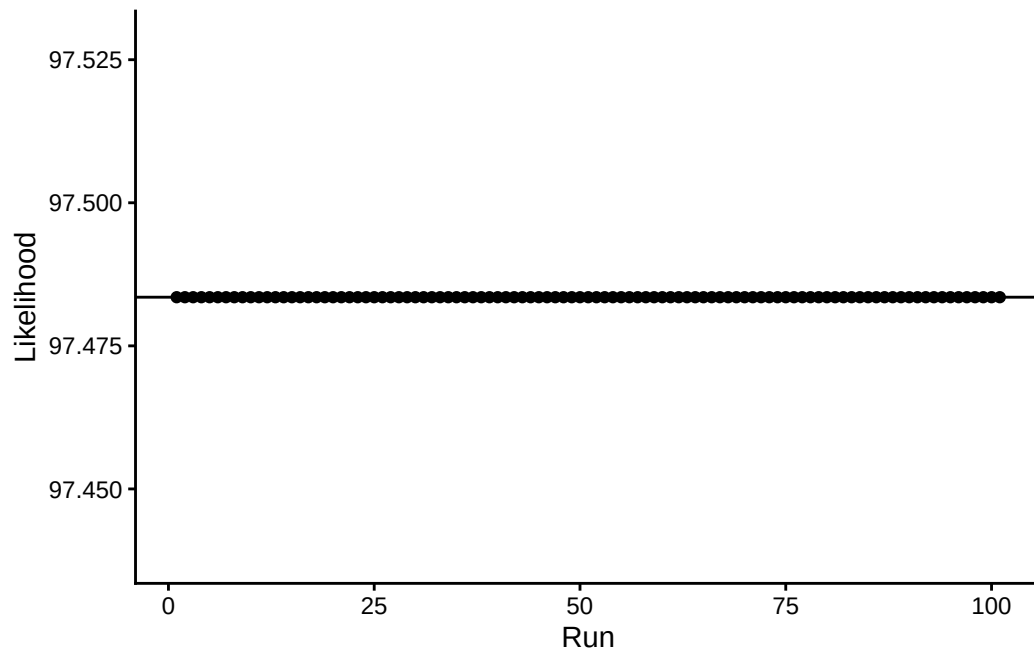


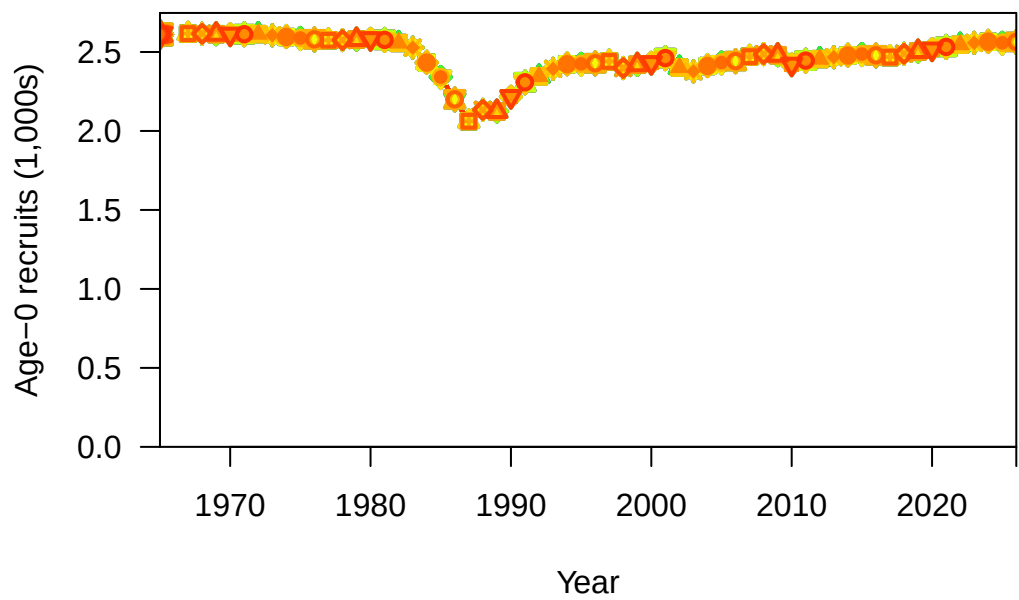
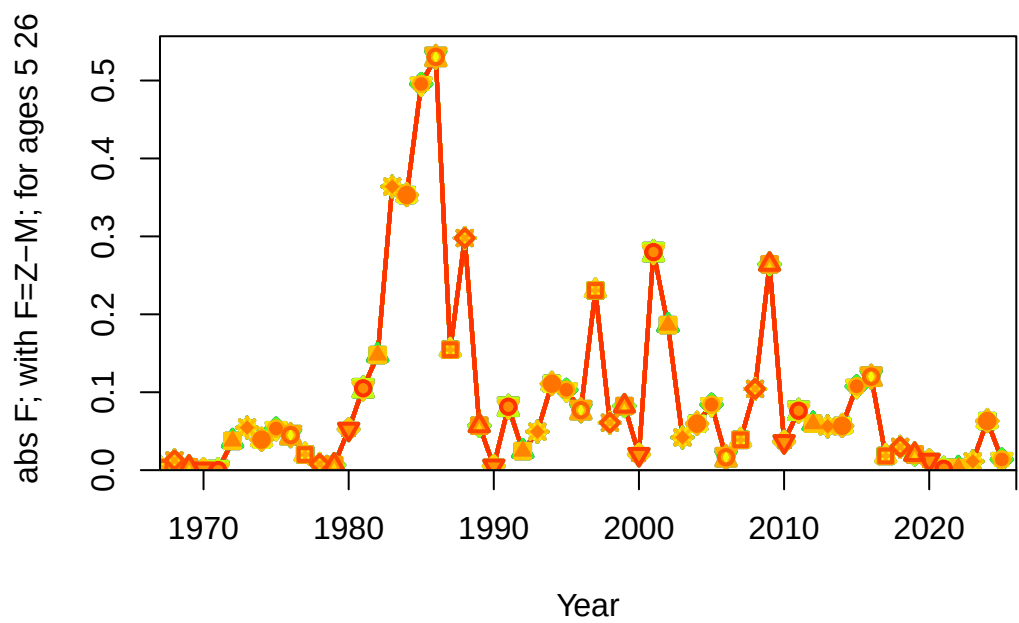




null device
1

Jitter





Selectivity and Maturity

